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Ammonia Emissions of Laying Hens as Affected by Stocking Density and Manure Accumulation Time

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Abstract. Different stocking densities (SDs) or space allocations are being practiced in commercial laying-hen operations as an attempt to improve hen welfare. Information concerning the impact of SD on accumulated manure properties (e.g., moisture content) and thus ammonia (NH₃) emissions is limited in the literature. Bird SD affects the amount of manure per unit of accumulated manure surface area, which may affect the NH₃ emission from the accumulated manure. A lab-scale study was conducted that resembled the conditions of manure-belt laying-hen houses for, with the objectives of (a) determining NH₃ emission rate (ER) of W36 laying hens housed under different SDs; (b) measuring the NH₃ emissions from laying hen manure during 7-d manure accumulation time (MAT); and (c) delineating the dynamics of feed disappearance, manure production and NH₃ ER during dark and light periods. Two different SDs were evaluated, 413 vs. 620 cm²/hen (64 vs. 96 in²/hen), designated as HD and LD, respectively. Ammonia ER was expressed in the units of NH₃ emission per bird, per kg of feed nitrogen (N) disappearance, per kg of as-is and dry manure, per kg egg output, per m² of projected manure surface area (PMA), and per animal unit (AU, 500 kg BW). Results show that SD effect on NH₃ ER is more pronounced for MAT ≥ 3d, where led to higher ER. Specifically, NH₃ emissions from the 3rd to 7th d MAT ranged from 41 to 307 mg/hen-d for HD and from 29 to 188 mg/hen-d for LD. This outcome supports the current industry practice of removing manure at 1- to 3-d MAT for the manure-belt house systems. Daily NH₃ ER increases exponentially with MAT (P<0.0001). The results also showed that hourly NH₃ ER during the night or dark period was as high as the hourly emissions during the daytime or light period.

Keywords: Hen welfare, floor area allocation, emission dynamics, manure accumulation

Introduction

Animal feeding operations (AFOs) are associated with aerial emissions, primarily ammonia (NH_3), nitrous oxide (N_2O), nitric oxide (NO), methane (CH_4), hydrogen sulfide (H_2S), volatile organic compounds (VOCs), and particulate matters (National Academy of Science, 2003). In agriculture, several sources of air emissions have been studied as well as the substances being emitted from them. Aerial emission rate (ER) is the product of source concentration of the substance and the air exchange rate through the source.

Among all constituents emitted from poultry production facilities, ammonia (NH_3) is the predominant noxious gas due to the nature of the manure (Liang et al., 2005). Livestock and poultry are often fed high protein diet, which contains surplus nitrogen, to ensure that the animals' nutritional requirements are met. Nitrogen that is not metabolized into animal protein or product is excreted as urea in the urine of cattle and swine or as uric acid from poultry where further microbial action releases ammonia into the air during manure decomposition (Gay et al., 2009). The ammonia volatilization rate from solid poultry manure is affected by nitrogen content, moisture content, stacking configuration of the manure pile, pH, temperature and oxygen availability, all of which contribute to the microbial activities and ammonia release from the manure pile (Li, 2006). Research has shown that prolonged exposure to high levels of NH_3 can cause reduced body weight gain and egg production in laying hens, and also can have a negative impact on farm workers (Carlile, 1984; Ning, 2008).

The most recent studies on NH_3 emissions from commercial U.S. poultry operations include those reported by Liang et al. (2005) for laying hens, Wheeler et al. (2006) and Burns et al. (2007) for broilers, and Li et al. (2008) for turkeys. Currently a national study through an air compliance agreement (ACA) between the U.S. EPA and certain sectors of the livestock and poultry industry is ongoing that aims to collect more baseline data on AFO air emissions. Laboratory experiments performed by Ning (2008) showed that NH_3 ER from laying hens (Hy-Line W-36) depends on duration of manure accumulation. Ning (2008) also found that the emissions had an inverse relation to defecation events. Moreover, hens may be housed at different stocking densities as producers respond to certain industry production management guidelines, such as those by the United Egg Producers and/or fast food-chain restaurants (e.g., McDonald's). Hence there is a need to quantify the impact of bird stocking density on aerial emission rate.

The objectives of this study were a) to quantify NH_3 emission rate (ER) of laying hens as affected by stocking density (SD) and manure accumulation time (MAT), and b) to delineate the dynamics of feed disappearance, manure production and NH_3 ER during dark and light periods. Results from this research will provide insight to the impact of production management practices on aerial emissions.

Material and Methods

Dynamic gas emission chambers system

The study was conducted using four dynamic gas emission chambers (fig. 1) at the Iowa State University Livestock Environment and Animal Physiology Laboratory II (LEAP Lab II). The chambers each had a dimension of 86 cm L x 45 cm W x 66 cm H and were located inside an environmentally controlled room. The chamber walls were constructed with transparent plexiglass panels (5 mm thick). Inside each chamber was an iron-framed wire-mesh cage (44 cm L x 34 cm W x 58 cm H). Fresh air to each chamber was supplied through an air distribution plenum to improve spatial uniformity, and the air supply was powered with a diaphragm air pump (100 L/min capacity, DDL 120-101, GAST Manufacturing INC, Benton Harbor, MI, USA)

placed on the inlet side of the chamber, thereby creating a positive-pressure ventilation system. Airflow rate through each chamber was measured with a thermoelectric air mass flow meter (capacity of 110 L/min, GFM57, Aalborg Instruments & Controls Inc., Orangeburg, NY, USA) placed in the supply air stream. Prior to onset of the experiment, calibration equations were developed to correlate output readings of the air mass flow meters with the actual flow rates. Air flow through each chamber was adjustable via a by-pass, so that the concentration of target gases (NH_3 , CO_2) inside the chamber could be controlled. One air temperature and relative humidity (RH) sensor (HMP45A/D, Vaisala, Woburn, MA, USA) was placed in each cage to measure the dry-bulb temperature. A plastic cup with tubing was placed underneath each nipple drinker to catch and divert any water leakage out of the manure pan or chamber.



Figure 1. ISU multi-chamber system for feeding, defecation and air emissions measurement.

To capture feeding and defecation events of the birds, two electronic balances (2200.0 ± 0.1 g, model GX2000, A&D Company Limited, Tokyo, Japan) with a 0-2.2 VDC analog output (sampling rate of 0.1 s, the data acquisition system averages data at every 10 s) were used in each chamber. One balance was used for measurement of the feeder weight or feeding activities and the other for measurement of the manure pan weight or defecation activities.

Samples of the exhaust air from each chamber were successively taken by a sampling pump (capacity of 20 L/min, Teflon wetted parts, Model No. 2107CA20B, Gardner Denver INC., Sheboygan, WI, USA) at 5 min intervals, with the first 3 min for stabilization and last 2 min for measurement. This sampling sequence yielded a measurement cycle of 25 min for the entire

system (including 5 min for the ambient air). The successive sampling was accomplished through controlled operation of five solenoid valves (PKV-2R-D1/4NF, Takasago Electric Inc., Midori-ku, Nagoya, Japan). A Teflon filter (4.5 cm diameter, 5 μ m pore diameter) connected to a Teflon tubing (1.63 cm diameter) was placed in front of each solenoid valve. A photoacoustic multi-gas analyzer (model 1412, INNOVA AirTech Instruments A/S, Ballerup, Denmark) was used to measure NH_3 and CO_2 concentrations. The multi-gas analyzer was challenged weekly and calibrated, as needed, with zero, 25 ppm NH_3 (balanced with air) span calibration and 2500 ppm CO_2 (N_2 balance) calibration gases. Dew-point temperature was measured with a dew-point hygrometer (model 2001, EG&G Moisture and humidity Systems, Burlington, MA).

Analog outputs from the temperature, INNOVA gas analyzer, dew-point hygrometer, electronic balances, and the mass flow meters were logged at 10 s intervals into a measurement and control module (CR10, Campbell Scientific, Inc., Logan, UT). All measurements were recorded as the average of output over the 10 s intervals.

To assess and ensure the integrity of the dynamic emission measurement system, CO_2 recovery tests with 100% ethanol ($\text{C}_2\text{H}_5\text{OH}$) lamps were conducted prior to the beginning of the experiment and repeated every other week, as performed by Ning (2008). In the test, an alcohol lamp containing 100% ethanol was placed on the manure pan electronic balance in each chamber, so that the dynamic as well as cumulative alcohol consumption could be obtained from the weight changes. Detailed algorithm for the recovery test was described by Scott & Hillman (1983).

The system was set and handled in a way that would mimic a manure-belt house system with a 7-d manure accumulation time period, so that the results of the study can be extrapolated to that housing type.

Hen handling and experimental design

The experimental Hy-Line W-36 hens were procured from a commercial farm in Iowa. Three batches of 10 randomly selected hens at initial age of 23 weeks were acquired. Each batch of 10 birds was randomly allocated to the four cages/chambers, two cages/chambers with 3 birds in each and the other two cages/chambers with 2 birds in each, thereby yielding two stocking densities, high stocking density (HD, 413 cm^2 or 64 in^2/hen) and low density (LD, 620 cm^2 or 96 in^2/hen). After 14 to 21-d measurement, the pullets inside the emission chambers were returned to the holding cages. Then, 10 laying hens from the next batch were allotted to the emission chambers to repeat the measurement. All the birds were kept at comfortable environmental conditions, as suggested by the Hy-Line Commercial Management Guide (i.e., 21.1-23.3°C, 40-50% relative humidity). Birds were weighed once a week.

During the test period, fresh feed was added daily to the feeder, usually between 10:00 and 12:00 h. Fluorescent lighting was provided at an illumination intensity of approximately 10 lux with a lighting program of 16L : 8D (i.e. 16 h light + 8 h dark). Nipple drinkers were used to supply drinking water. Manure pans were replaced after 7-d manure accumulation (1 cycle). Prior to the onset of each trial, the cages/chambers were cleaned and disinfected, so that the first cycle started with 0 (zero) NH_3 emissions in the system. For the following cycles of the same trial, NH_3 from the previous cycle was still being flushed out of the system during the first day of manure accumulation. Residual NH_3 from the previous cycle was presumably adsorbed by the interior surfaces of the chambers (walls, cage, and bird's body surface), which would represent production situation.

To complete the randomization process and avoid chamber effect on measurements, groups under the same treatment switched chambers on a weekly basis, so that by the end of the trial, all SDs would have been run in all four chambers.

Because the hens used in this research study ranged from 23 to 32 weeks of age, a considerably wide age span, data sets were first analyzed for age effect. However, no age effect was detected on feed disappearance, manure production rates or NH₃ ER. Consequently, the age factor was disregarded and the data were pooled over the age span.

Analyzed variables

Calculation of NH₃ ER and evaluation of SD effect on the NH₃ emissions

Daily ammonia emission rate (NH₃ ER) was calculated for 1 to 7 d of MAT with the following equation.

$$NH_3 \text{ ER} = \frac{Q_{STPD}(C_{NH_3,e} - C_{NH_3,i}) \times W_m \times 10^3}{10^6 \times V_m \times N} \quad [mg \cdot h^{-1} \cdot hen^{-1}] \quad [1]$$

Where:

Q_{STPD} – air flow rate, corrected for standard temperature (21° C), pressure (1 ATM) and dry basis, L·h⁻¹·chamber⁻¹

$C_{NH_3,e}$, $C_{NH_3,i}$ – exhaust and inlet ammonia concentrations, respectively, ppm

W_m – molecular weight of ammonia (17.031 g·mol⁻¹)

V_m – molar volume of ammonia, corrected for standard temperature (21° C), pressure (1 ATM), 24.14 L·mol⁻¹

N – number of hens per cage/chamber, 2 or 3

NH₃ ER was calculated in several units, including g/kg N disappearance, g/kg ‘as is’ and ‘dry basis’ manure, g/kg egg, and g/AU (animal unit, 500 kg BW). The feed N disappearance was calculated based on the feed disappearance and crude-protein (CP) content of the diet. Crude protein was divided by 6.25 to yield the feed N content. Effects of SD were tested on a daily basis using *proc glm* in SAS. In addition, the percent of the difference in the mean ER values between the SD densities was calculated using equations 2 and 3. Standard error values for the percent of the difference were calculated using the Delta Method, as described by Casella & Berger (2002).

$$\hat{\lambda} = \left(\frac{\hat{\mu}_{LD} - \hat{\mu}_{HD}}{\hat{\mu}_{HD}} \right) \times 100 \quad [2]$$

Where:

$\hat{\lambda}$ - estimated mean value of the percent of the difference (%)

$\hat{\mu}_{LD}$ - estimated mean value for the variable μ under the LD treatment

$\hat{\mu}_{HD}$ - estimated mean value for the variable μ under the HD treatment

$$SE^2(\hat{\lambda}) = \begin{bmatrix} \frac{\partial \lambda}{\partial \mu_{HD}} & \frac{\partial \lambda}{\partial \mu_{LD}} \end{bmatrix}_{(\mu_{HD}, \mu_{LD})=(\hat{\mu}_{HD}, \hat{\mu}_{LD})} \begin{bmatrix} \hat{\text{var}}(\hat{\mu}_{HD}) & \hat{\text{cov}}(\hat{\mu}_{HD}, \hat{\mu}_{LD}) \\ \hat{\text{cov}}(\hat{\mu}_{HD}, \hat{\mu}_{LD}) & \hat{\text{var}}(\hat{\mu}_{LD}) \end{bmatrix} \begin{bmatrix} \frac{\partial \lambda}{\partial \mu_{HD}} \\ \frac{\partial \lambda}{\partial \mu_{LD}} \end{bmatrix}_{(\mu_{HD}, \mu_{LD})=(\hat{\mu}_{HD}, \hat{\mu}_{LD})} \quad [3]$$

Where:

$SE^2(\hat{\lambda})$ - Standard error of the estimated percent of the difference;

$\hat{\text{var}}(\hat{\mu}_{HD})$ - Estimated variance of the estimated mean under the HD treatment, obtained from SAS output;

$\hat{\text{var}}(\hat{\mu}_{LD})$ - Estimated variance of the estimated mean under the LD treatment, obtained from SAS output;

$\hat{\text{cov}}(\hat{\mu}_{HD}, \hat{\mu}_{LD})$ - Estimated covariance for the estimated means under the effects of both treatments (equal to zero, since the number of replicates per SD treatment is always the same).

Capturing the dynamics of feed disappearance, manure production and NH₃ emissions throughout light and dark periods

Hourly feed disappearance and fresh manure production rates were calculated from the continuously measured feeder and manure pan weights. The total values of feed disappearance, manure production and NH₃ emission were broken down into light and dark periods, for each of the 7 d MAT.

To estimate the amount of fresh manure produced by the hens (accounting for moisture evaporation), an algorithm was written using Macros in Excel. Figure 2 shows the steps of the algorithm. Manure weight data, originally recorded at 10 s intervals were averaged into 1-min intervals to reduce the scale noise. Then the change (delta) in manure weight after every minute was calculated, with negative deltas considered as moisture loss and positive delta indicating occurrence of a defecation event. Adding up the positive and negative deltas provided 'as-is' manure production. All variables were initially calculated in g/hen-h.

Analysis of the Projected Manure Area (PMA)

Because NH₃ ER also relates to the amount of manure surface area that is exposed to the surrounding air, pictures of the undisturbed manure pans were taken at the 7th day of manure accumulation (fig. 3). The images were then imported into AutoCAD and had their projected manure area (PMA) estimated. PMA data were then related to manure weight and daily NH₃ ER at the 7th d of MAT.

All variables presented were tested for SD effect using *proc glm* in SAS through the ANOVA for a block design. Significant difference between means under different SD was tested through the Tukey Test and p-values less or equal to 0.05 were considered significant.

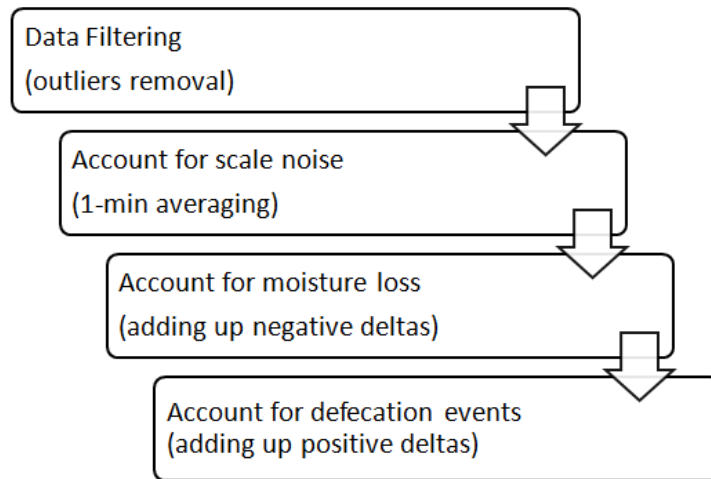


Figure 2. Flow chart illustrating the steps of the algorithm for manure production data processing.

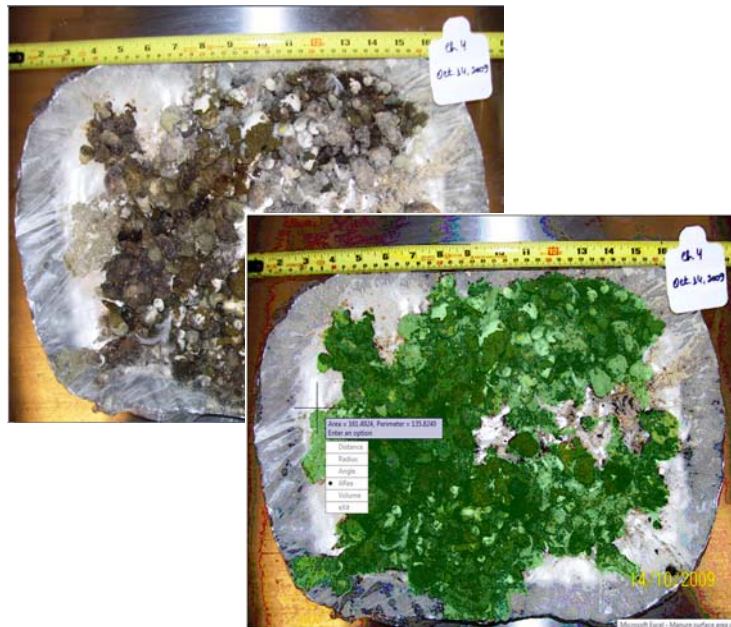


Figure 3. Pictorial sample of a projected manure area (PMA) calculation using the 'area calculator' tool in AutoCAD. Original RGB image (top) and image imported into AutoCAD desktop (bottom) with polygon highlighted in green and value for total PMA calculated in u^2 .

Results and Discussion

Effects of SD and MAT on feed disappearance, manure production and ammonia emissions

From figure 4, one can see the difference in NH_3 emissions at the beginning of the cycle for a 'clean system' (top) vs. the 'non-clean system' (bottom) for a period of 7-d of MAT (168 h). For data analysis purposes, a distinction was made between 'clean' vs. 'non-clean' system and NH_3 ER in several units were derived from the daily NH_3 ER obtained from 'clean' system. The number of replicates was 4 for the 'clean system' situation and 12 for the 'non-clean system' situation. Figure 5 displays the difference between HD and LD for both 'clean' and 'non-clean' systems. One can see that for the 'non-clean' system, most of the residual ammonia is flushed before the first 48-h of MAT. In addition, it is important to note that both curves seem to remain parallel to each other after 2-d MAT. Also, the difference in NH_3 ER between the HD and LD regimens increased with MAT until approximately 4-d MAT (96 h), after which the difference remained by and large unchanged.

Data on feed and feed N disappearance, 'as is' and dry manure weight, egg weight and NH_3 ER in various units over the 7-d MAT are summarized in Table 1. There was no significant difference ($P = 0.46 - 0.60$) in feed disappearance between the two SD regimens for all MATs, 98 ± 2 and 99 ± 2 g/hen for HD and LD, respectively. The feed disappearance values agree with what is presented in the guidelines for W-36 adult layers (100 g/hen-d). Egg weight was also not affected by SD ($P = 0.7 - 0.9$), 58.5 ± 0.3 and 58.6 ± 0.3 g/egg for HD and LD, respectively. These outcomes indicate that the reduced floor space allocation did not adversely affect feed use or egg production.

Mean daily NH_3 ER data presented in table 1 for 'clean' and 'non-clean' systems look different from each other for the first 2-d MAT; however after the 3rd day the 95% confidence intervals for both conditions overlap each other. When looking at SD effect on NH_3 ER for 'clean' vs. 'non-clean' systems, one can observe that percent of the difference values are consistently negative and relatively constant for $\text{MAT} \geq 3$ d, evidencing that the treatment HD led to higher NH_3 ERs, and with an overall difference of -35 ± 20 and -29 ± 10 % for 'clean' and 'non-clean' systems respectively. This outcome indicates that the SD will likely impact the emission after 3-d MAT. In particular, significant effect of SD was detected at the 7th d of MAT for the 'clean' system, 307 ± 30 mg/hen-d for HD vs. 188 ± 30 mg/hen-d for LD; and at days 3 and 4 of MAT for the 'non-clean' system, with estimated means being 45 ± 3 mg/hen-d for HD vs. 25 ± 3 mg/hen-d for LD at 3-d MAT and 83 ± 8 mg/hen-d for HD vs. 56 ± 8 mg/hen-d for LD at 4-d MAT. The analysis of the plot of residuals for NH_3 ER data indicated that there is an increase in the uncertainty of the emission as the estimated mean became larger. Thus a greater number of replicates would presumably allow one to see more significant effects on daily NH_3 ER for both 'clean' and 'non-clean' systems after the 3rd d of MAT.

This outcome on daily NH_3 ER supports current management practices used in manure-belt housing systems, where manure is usually removed every 1 to 3 d MAT to avoid overload of the belts (Xin, 2010 – Personal communication). Results indicate that regardless of the stocking density, daily NH_3 ER will increase considerably for $\text{MAT} > 3$ d.

Liang et al. (2005) measured NH_3 ER from manure-belt laying hen houses with $\text{MAT} = 1$ -d (Iowa) or 3- to 4-d (Pennsylvania) and reported that the overall NH_3 ER was 54 ± 5 mg/hen-d for $\text{MAT} = 1$ d and 94 ± 2 g/hen-d for $\text{MAT} = 4$ d. These ER values parallel those in the current study for $\text{MAT} = 2$ d of the non-clean HD system (57 ± 6 mg/hen-d) and 3-4 d of the clean or non-clean HD system (83-94 mg/hen-d). The 2-d MAT of the non-clean system in the current study likely better resembles the commercial production situation in that some residual manure exists in the barn from the daily removal of the manure.

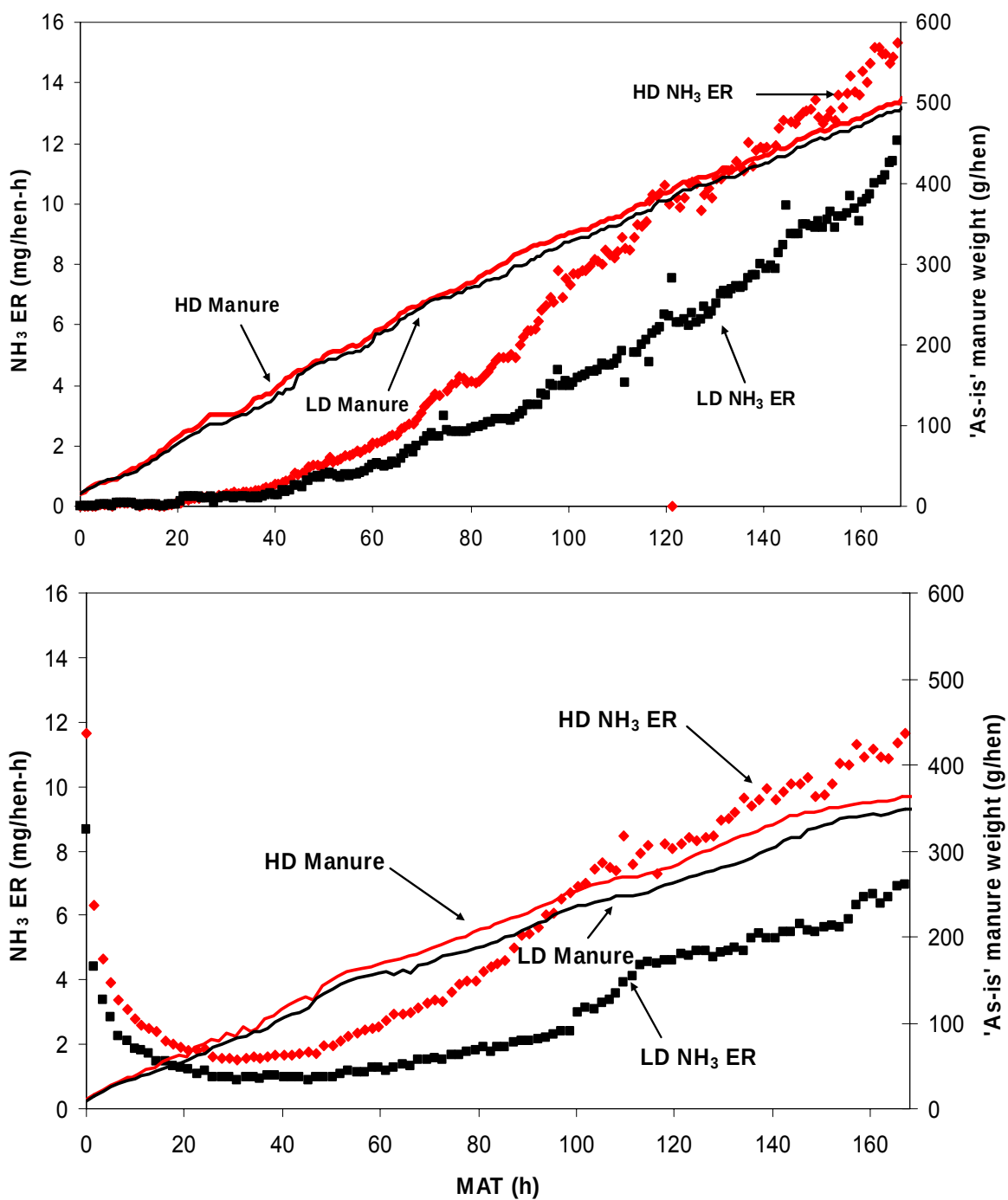


Figure 4. Daily NH_3 ER and 'as is' manure production profiles for a 7-d manure accumulation time (MAT) for clean system (top) and non-clean system (bottom) and two stocking densities of high density (HD = 413 cm^2/hen) and low density (LD = 620 cm^2/hen).

Ammonia ER in other units followed similar trends to that of ER in g/hen-d (table 1). The overall percent of difference in grams of NH_3 emissions per N disappearance was -35 ± 23 %, and the SD effect became significant at 4- and 6-d MAT.

Stocking density effects on NH_3 emissions in g/kg manure were similar for manure expressed in 'as-is' and dry basis for most days of MAT. A significant effect of SD on NH_3 ER on a 'as-is' basis was detected at 7-d MAT. Ammonia emissions under the treatment LD was 27 ± 16 % lower than that for the treatment HD in g/kg 'as-is' manure, and 31 ± 19 % in g/kg dry manure.

Ammonia emissions in g/kg egg and g/AU-d were both affected by SD only on 7-d MAT, being 5.2 ± 0.5 g/kg egg for HD and 3.2 ± 0.5 g/kg egg for LD; and 101 ± 9 g/AU-d for HD and 60 ± 9 g/AU-d for LD.

Light vs. dark period dynamics of hen feed disappearance, manure production and NH_3 emissions

One can see from figure 6 (top) how hourly 'as-is', fresh manure production rates and hourly moisture loss rates are related. The algorithm was validated by correlating the daily 'as-is' manure production obtained from the difference between beginning and end of the day (called raw data method) with the daily 'as-is' manure production obtained from the algorithm by adding up the hourly rates for the whole 24-h period. The degree of correlation is quite good, as presented in figure 6 (bottom) ($R^2=0.96$).

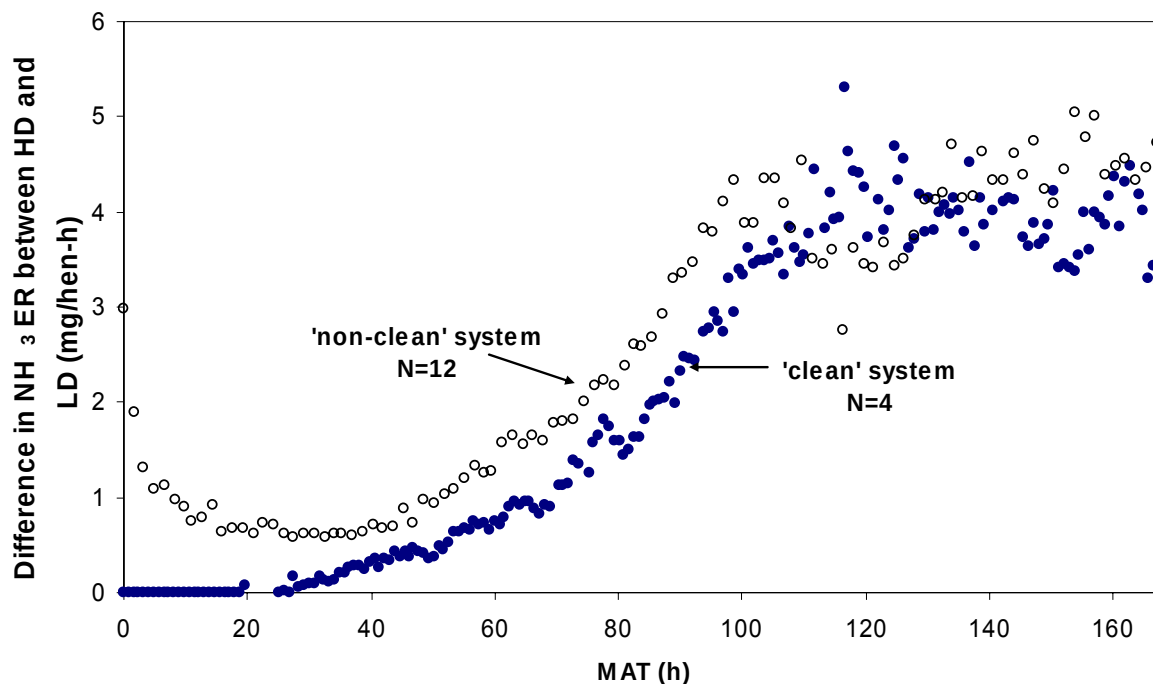


Figure 5. Difference in NH_3 ER between high density (HD = $413 \text{ cm}^2/\text{hen}$) and low density (LD = $620 \text{ cm}^2/\text{hen}$) for 'clean' and 'non-clean' systems (MAT = manure accumulation time).

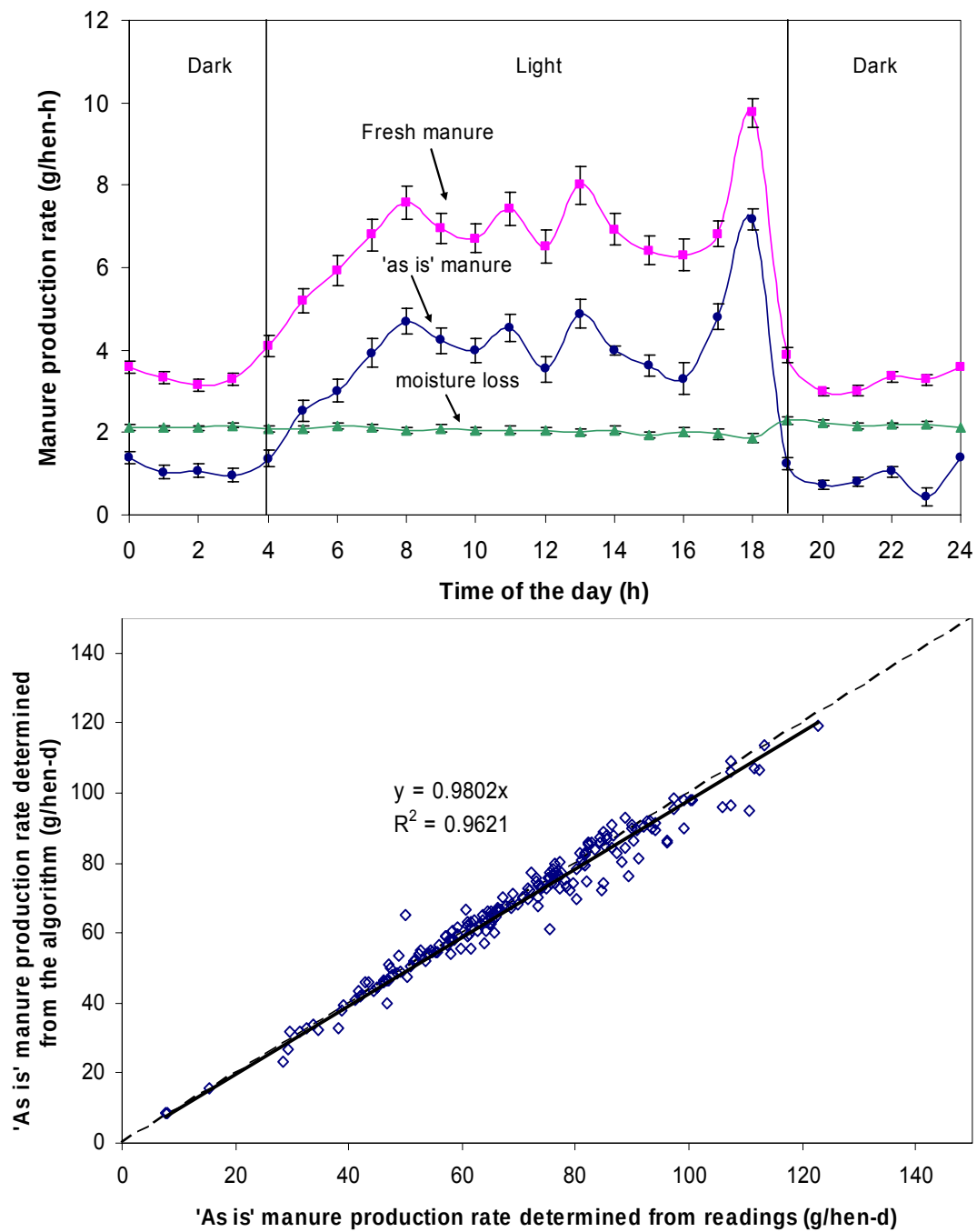


Figure 6. Dynamics of fresh, 'as is' manure production rates and moisture loss from manure throughout dark and light periods (top) and correlation between 'as is' manure production obtained from the algorithm and calculated from manure weight readings (bottom).

Feed disappearance and fresh manure production during dark and light periods are shown in figure 7 and table 2. Figure 7 shows that during the dark period, the hen did not eat; however, a small amount of manure was still being produced over the dark hours.

As shown by the data in table 2, the overall fresh manure production during the light period of the day was 109 ± 6 g/hen for HD and 105 ± 6 g/hen for LD, ($P = 0.70 - 0.11$). The overall mean manure production during dark period was 28 ± 1 g/hen and 24 ± 1 g/hen for HD and LD, respectively.

The data in figure 8 show the relationship between feed disappearance and fresh manure production rates. A linear model was fit to the data set ($R^2=0.72$), which suggested that the manure production increased by 0.5 g/g of feed disappearance.

One can observe from daily NH_3 ER data presented in table 2 (for both clean and non-clean systems), that considering the fact that the light period was twice as long as the dark period (16 h L vs. 8 h D), the emissions for HD and LD in the dark period would be relatively equal to the respective emissions for the light period if light and dark hours had the same number of hours. This fact is an indication that emissions tend to occur during the dark periods. It is speculated that during the day, 'fresh' manure being produced is constantly covering the 'old' manure which is more likely to emit NH_3 than the 'fresh' manure. Since the fresh manure production rate is considerably lower during the night, more NH_3 would emit from the 'old' manure.

This outcome indicates that if an NH_3 -suppressing agent is applied to the hen manure, it would be more effective to apply the agent during the dark period.

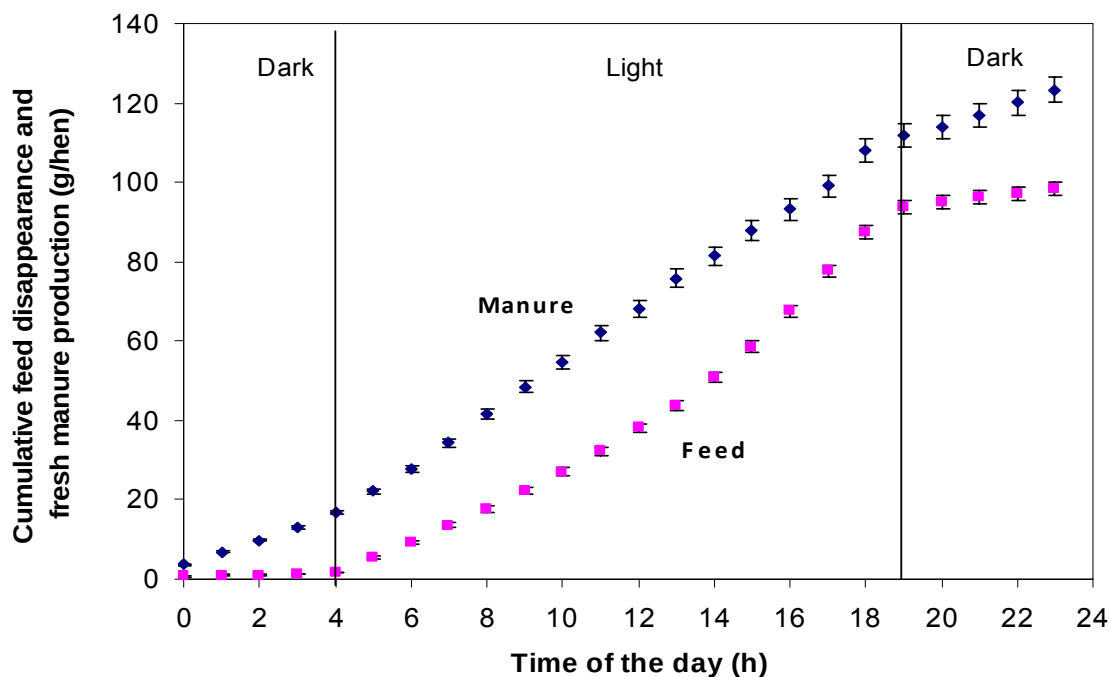


Figure 7. Cumulative fresh manure production and feed disappearance in a 24-h period. (Laying hen age: 23-34 wk)

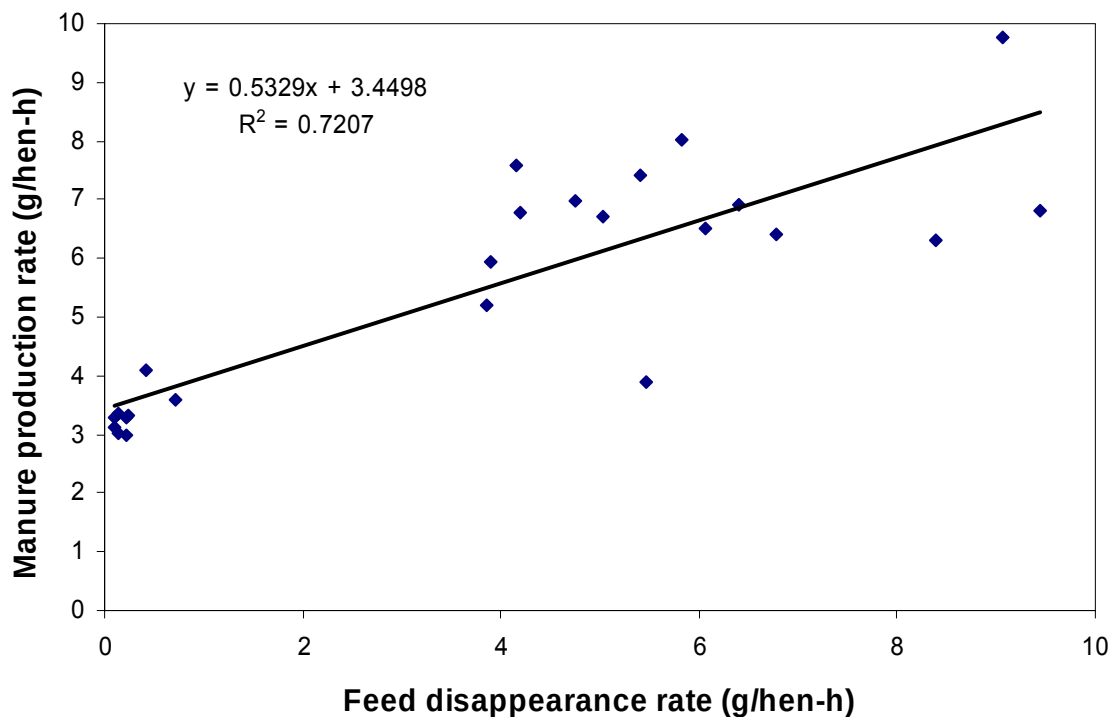


Figure 8. Linear relationship between feed disappearance and fresh manure production for laying hens. (laying hen age: 23-34 wk)

Analysis of the Projected Manure Area (PMA) data

The PMA values in cm²/hen-d for the 7th d of MAT are shown in table 3. PMA was higher for the LD treatment than for the HD, 273±11 vs. 312±11 cm²/hen-d (P = 0.04). The surface area to manure weight ratio was also significantly higher for LD than for HD, 0.65±0.03 vs. 0.55±0.03 cm²/g (P = 0.05). LD treatment had significantly lower NH₃ ER in g/m²-d, 9±1 vs. 15±1 g/d-m² PMA (P = 0.003).

Regression of Daily NH₃ ER to SD and MAT

A regression analysis was performed on the daily NH₃ ER data, as a function of SD (in mg/hen-d) and MAT (d). A log transformation was applied to the data set to account for the augmented SE of the mean as the mean increased. Results from the statistical analysis indicated that NH₃ ER could be explained as a function of SD and MAT following an exponential fashion (P<0.0001). From eq. 4 one can see that daily NH₃ ER is negatively correlated to the amount of floor area allocation, in other words, a decrease in the floor area allocation (or higher SD) will cause an increase in daily NH₃ ER, and vice versa. However, as it was expected, the daily NH₃ ER was positively correlated to MAT.

$$NH_3ER = e^{[(2.84 \pm 0.33) - (0.0020 \pm 0.0004)SD + (0.597 \pm 0.045)MAT]} \quad (R^2 = 0.81) \quad [4]$$

Where:

NH₃ ER – NH₃ emission rate (mg hen⁻¹ d⁻¹)

SD – stocking density (cm² hen⁻¹)

MAT – manure accumulation time (d)

Conclusions

Effects of stocking density (SD) and manure accumulation time (MAT) of laying hens on ammonia emission were examined. Dynamics of feed use, defecation and NH₃ emission were also characterized. The following conclusions were drawn:

1. SD effect on NH₃ emission became more pronounced for MAT ≥ 3 d.
2. Daily NH₃ emission increases with MAT, but tends to decrease with increasing floor space allocation to the hens.
3. Ammonia emissions from the manure during dark hours tend to be higher than during light hours.
4. Hens do not eat during dark hours, but 20% of the total daily manure is produced during the night or dark hours. The relationship of manure production and feed disappearance follow a linear function.

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Table 1. Cumulative feed disappearance, feed nitrogen (N) disappearance, manure weight and ammonia emission of W-36 hens over 7-day period at two cage stocking densities (SD): hen age = 23 - 37 wk; hen body weight = 1313-1987 g; HD = 413 cm²/hen; LD = 620 cm²/hen. NH₃ emissions in g/kg N disappearance, g/kg 'as-is' and dry manure, g/kg egg output and g/AU-d were calculated based on 'clean' system NH₃ ER data (mean ± SE).

Variables		Manure Accumulation Time (MAT, d)							
		1	2	3	4	5	6	7	Overall
Feed disappearance, g/bird-d	HD	94±2	98±3	99±3	96±8	103±4	98±2	99±3	98±2
	LD	95±2	100±3	98±3	100±8	102±4	99±2	98±3	99±2
Feed N disappearance, g/bird-d	HD	2.60±0.07	2.72±0.08	2.73±0.08	2.63±0.08	2.83±0.09	2.70±0.05	2.75±0.07	2.71±0.06
	LD	2.62±0.07	2.75±0.08	2.73±0.08	2.77±0.08	2.80±0.09	2.72±0.05	2.71±0.07	2.73±0.06
Egg weight (g/egg)	HD	59±1	59±1	57.6±0.3	57±1	59.1±0.4	59±1	59±1	58.5±0.3
	LD	58±1	59±1	58.6±0.3	58±1	58.2±0.4	59±1	59±1	58.6±0.3
Manure weight (as is), g/bird	HD	45±4	137±5	225±9	307±12	375±18	455±18	425±18	-
	LD	46±4	134±5	214±9	288±12	360±18	419±18	393±18	-
Manure weight (dry basis), g/bird	HD	26±2	72±6	130±5	177±7	216±10	262±10	245±10	-
	LD	26±2	80±6	127±5	171±7	214±10	248±10	234±10	-
NH ₃ emission, mg/bird-d (clean system)	HD	5±3	12±4	41±9	98±13	179±26	251±33	307±30 a	-
	LD	7±3	12±4	29±9	64±13	114±26	160±33	188±30 b	-
	%diff	-	1±50	-31±33	-34±22	-36±24	-36±22	-39±17	-35±20
NH ₃ emission, mg/bird-d (non-clean system)	HD	349±29	57±6	45±3 a	83±8a	154±19	245±27	337±37	-
	LD	260±29	38±6	25±3 b	56±8b	107±19	171±27	247±37	-
	%diff	-	-33±17	-43±14	-32±16	-30±19	-30±17	-26±17	-29±10
NH ₃ emission, g/kg N disappearance	HD	2±1	4±1	16±3	40±4 a	61±11	105±11 a	115±12	-
	LD	2±1	4±1	11±3	24±4 b	42±11	62±11 b	69±12	-
	%diff	-	-	-35±30	-38±18	-32±29	-40±18	-31±20	-35±23
NH ₃ emission, g/kg manure (as is)	HD	0.07±0.05	0.1±0.02	0.15±0.02	0.30±0.03	0.42±0.04	0.52±0.06	0.58±0.05a	-
	LD	0.07±0.05	0.1±0.02	0.12±0.02	0.22±0.03	0.30±0.04	0.35±0.06	0.38±0.05b	-
	%diff	-	-	-17±17	-25±15	-29±16	-33±19	-35±14	-27±16
NH ₃ emission, g/kg manure (dry basis)	HD	0.10±0.07	0.12±0.04	0.27±0.04	0.52±0.05	0.8±0.1	0.9±0.1	1.0±0.1	-
	LD	0.15±0.07	0.10±0.04	0.20±0.04	0.35±0.05	0.5±0.1	0.6±0.1	0.7±0.1	-
	%diff	-	-	-27±24	-33±14	-30±21	-32±17	-35±14	-31±19
NH ₃ emission, g/kg egg	HD	0.08±0.07	0.20±0.08	0.7±0.1	1.6±0.2	3.0±0.4	4.2±0.6	5.2±0.5 a	-
	LD	0.10±0.07	0.20±0.08	0.5±0.1	1.1±0.2	2.0±0.4	2.7±0.6	3.2±0.5 b	-
	%diff	-	-	-32±32	-33±20	-33±23	-37±24	-39±16	-35±19
NH ₃ emission, g/AU-d	HD	2±1	4±1	14±3	32±4	58±8	82±10	101±9 a	-
	LD	2±1	4±1	9±3	20±4	36±8	51±10	60±9 b	-
	%diff	-	-	-34±32	-37±20	-38±23	-37±7	-40±15	-37±18

Values for the two stocking densities of each variable followed by different letters are significantly different (**a** and **b** for 0.01 < P ≤ 0.05, and **x** and **y** for P ≤ 0.01; AU = animal unit = 500 kg live body weight; %diff ($\hat{\lambda}$) values were calculated from: $\hat{\lambda} = [(\hat{\mu}_{LD} - \hat{\mu}_{HD}) / \hat{\mu}_{HD}] \cdot 100\%$, where $\hat{\mu}_{LD}$ and $\hat{\mu}_{HD}$ are the estimated means for the considered variable under low and high density, respectively. The standard error for $\hat{\lambda}$ was estimated with the Delta Method. However, when the %diff was small, the Delta Method tended to output extremely high SEs, in other words, the function used to estimate SE got closer to its limit range ($\lim_{\%diff \rightarrow 0} SE = \infty$), thus %diff values with extremely magnified estimates for SE were deleted.

Table 2. Estimated means and SEs from the two-way ANOVA test for a block design of: fresh manure production, feed disappearance and NH₃ ER for 'clean' and 'non-clean' system during the light, dark and daily periods (16 L vs. 8 D) along the seven days of MAT of W-36 hens at and two cage stocking densities (SD): hen age = 23-34 wk; hen body weight = 1313 (20 hens) – 1687 (10 hens) g; HD = 413 cm²/bird; LD = 620 cm²/bird.

Variable	Period	SD	Manure accumulation time (MAT-d)							
			1	2	3	4	5	6	7	Overall
Fresh manure weight (g/hen-period)	Light	HD	100±8	127±17	131±16	120±14	97±5	92±3	93±5	109±6
		LD	107±8	127±17	98±16	108±14	103±5	95±3	103±5	105±6
	Dark	HD	29±2	29±1 a	31±2	29±1 a	28±1 a	26±1	27±1	28±1 a
		LD	24±2	24±1 b	24±2	25±1 b	24±1 b	25±1	24±1	24±1 b
	Daily	HD	129±8	156±17	162±18	149±15	124±5	118±3	120±4	137±7
		LD	131±8	150±17	122±18	132±15	128±5	120±3	127±4	130±7
Feed disappearance (g/hen-period)	Light	HD	93±1	97±2	98±3	95±7	100±4	97±1	97±2	96±2
		LD	93±1	95±2	97±3	99±7	101±4	96±1	97±2	97±2
	Dark	HD	1±1	1±1	1.1±0.3	0.9±0.3	3.1±0.4 a	1±1	2±1	1.3±0.4
		LD	2±1	3±1	1.1±0.3	1.2±0.3	1.1±0.4 b	3±1	1±1	1.9±0.4
	Daily	HD	94±2	98±3	99±3	96±8	103±4	98±2	99±3	98±2
		LD	95±2	100±3	98±3	100±8	102±4	99±2	98±3	99±2
Daily NH ₃ ER (mg/hen-period) (clean system)	Light	HD	3±2	8±3	25±5	61±7	108±18	168±21	203±20 a	-
		LD	4±2	7±3	18±5	39±7	72±18	102±21	112±20 b	-
	Dark	HD	2±1	4±1	16±4	37±5	71±7 a	83±12	104±11	-
		LD	3±1	5±1	11±4	25±5	42±7 b	58±12	76±11	-
	Daily	HD	5±3	12±4	41±9	98±13	179±26	251±33	307±30 a	-
		LD	7±3	12±4	29±9	64±13	114±26	160±33	188±30 b	-
Daily NH ₃ ER (mg/hen-period) (non-clean system)	Light	HD	212±17	36±4	28±2 a	51±5 a	88±13	158±19	206±26	-
		LD	165±17	22±4	14±2 b	33±5 b	63±13	99±19	149±26	-
	Dark	HD	137±12	21±2	17±1 a	32±4	67±6 a	87±8	131±11	-
		LD	95±12	16±2	11±1 b	23±4	44±6 b	72±8	98±11	-
	Daily	HD	349±29	57±6	45±3 a	83±8a	154±19	245±27	337±37	-
		LD	260±29	38±6	25±3 b	56±8b	107±19	171±27	247±37	-

Values for the two stocking densities of each variable followed by different letters are significantly different (**a** and **b** for 0.01 < P ≤ 0.05, and **x** and **y** for P ≤ 0.01).

Table 3. Estimated mean and SE and p values from the two-way ANOVA for a block design of: manure moisture content, projected manure area (PMA), PMA to manure wt. ratio, manure weight, NH₃ emission rate (ER), NH₃ ER to PMA ratio at the 7th d manure accumulation time (MAT) of W-36 hens at and two cage stocking densities (SD). Hen age = 23-34 wk; hen body weight = 1313 (20 hens) – 1687 (10 hens) g; HD = 413 cm²/bird; LD = 620 cm²/bird, N=5 observations for all variables at both SDs.

Variable	SD	Mean±SE	P value	
			SD	Block
Manure moisture content (% w.b.)	HD	73±1 x	0.007	0.003
	LD	69±1 y		
PMA (cm ² /hen-d)	HD	273±11b	0.04	0.18
	LD	312±11a		
Area/manure wt. ratio (cm ² /g)	HD	0.55±0.03 b	0.05	0.008
	LD	0.65±0.03 a		
‘As is’ manure wt. (g/hen) over 7 d MAT	HD	518±20	0.19	0.04
	LD	478±20		
NH ₃ ER (g/hen-d) over 7 d MAT	HD	0.46±0.04 a	0.01	0.76
	LD	0.29±0.04 b		
NH ₃ ER (g/m ² -d)	HD	15±1y	0.003	0.04
	LD	9±1x		

Values for the two stocking densities of each variable followed by different letters are significantly different (**a** and **b** for 0.01 < P ≤ 0.05, and **x** and **y** for P ≤ 0.01).